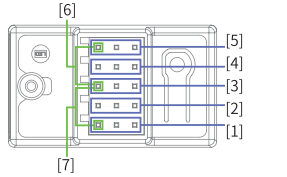
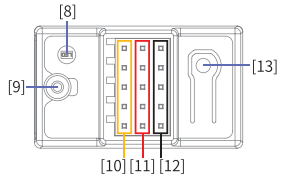
产品介绍 Introduction

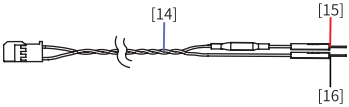
Ar4 采用 Antelope 协议（羚羊版自动跳频数字系统），是一款 4 通道超低延时接收机。它具有外置单天线，支持单双向传输和回传功能，设计小巧轻便，易于安装，支持 4 个 Newport 功能接口 (Newport 接口可自定义输入或输出的信号类型: i-BUS2/S.BUS/i-BUS/PPM/PWM 等)。适配教育、竞技、娱乐模型使用。

The Ar4 adopts the Antelope Protocol (Antelope Automatic Frequency-Hopping Digital System), a 4-channel ultra-low-latency receiver. It features an external single antenna and supports 1-way/2-way transmission and telemetry. Its compact, lightweight design enables easy installation. It supports 4 Newport interfaces (Newport interface can be customized with input or output signal types: i-BUS2/S.BUS/i-BUS/PPM/PWM, etc.). Compatible with educational, competitive, and recreational models.

接收机概览 Receiver Overview







[1] CH1(NPD)
[2] CH2(NPC)
[3] CH3(NPB)
[4] CH4(NPA)
[5] BVD/VCC（电池电压检测 / 供电接口）
[6] 对码接口
[7] 强制更新接口
[8] LED 灯
[9] 天线
[10] 信号端
[11] +（电源正极）
[12] -（电源负极）
[13] 对码按键
[14] FS-XC101 电池电压检测线
[15] 接电池正极
[16] 接电池负极

[1] CH1(NPD)
[2] CH2(NPC)
[3] CH3(NPB)
[4] CH4(NPA)
[5] BVD/VCC(Battery voltage detection/ Power supply interface)
[6] BIND Interface
[7] Forced Update Interface
[8] LED
[9] Antenna
[10] S (Signal Pin)
[11] + (Power anode)
[12] - (Power cathode)
[13] BIND Button
[14] FS-XC101 battery voltage detection cable
[15] Connect to battery anode
[16] Connect to battery cathode

注：BVD 电压检测范围： 0~70V

Note: BVD voltage detection range: 0~70V

产品规格 Product Specifications

- 产品型号：Ar4
- 适配发射机：支持 Antelope 协议的发射机
- 适配模型：教育、竞技、娱乐模型
- PWM 通道数：4
- 无线频率：2.4GHz ISM
- 无线协议：Antelope
- 发射功率：<20dBm
- 天线类型：单天线
- 工作电压：3.5 ～ 9V/DC
- 数据输出：i-BUS2/S.BUS/i-BUS/PPM/PWM
- 温度范围：-10°C ~ +60°C
- 湿度范围：20%~95%
- 固件更新：支持
- 通道分辨率：4096 级
- 外形尺寸：29*17*16.6mm
- 防水等级：PPX6
- 机身重量：6.2 g
- 认证 : CE、 FCC ID: 2A2UNAR400

- Product Model: Ar4
- Compatible Transmitters: All transmitters with Antelope protocol
- Compatible RC Models: Educational,competition, and recreational models
- Number of PWM Channels: 4
- RF: 2.4GHz ISM
- RF Protocol: Antelope
- Maximum Power: <20dBm (e.i.rp.) (EU)
- Antenna: Single Antenna
- Operating Voltage: 3.5~9V/ DC
- Data Output: i-BUS2/S.BUS/i-BUS/PPM/PWM
- Temperature Range: -10°C ~+60°C
- Humidity Range: 20%~95%
- Firmware Update: Supported
- Resolution: 4096
- Dimensions: 29*17*16.6mm
- Waterproof: PPX6
- Weight: 6.2 g
- Certifications: CE, FCC ID: 2A2UNAR400

对码 Binding

Ar4 接收机支持双向对码和单向对码。在双向对码完成后，发射机会显示接收机回传的信息。因此，对码之前，需要在发射机端先设置好是单向对码还是双向对码。如需对码接收机与发射机，请按照以下步骤操作：

双向对码步骤：

1. 发射机选择双向通信，然后进入对码状态；

The Ar4 receiver supports both 2-way and 1-way binding. After completing 2-way binding, the transmitter will display the information returned by the receiver. Therefore, before binding, it is necessary to set whether it is 1-way binding or 2-way binding on the transmitter side. If you need to bind the receiver with the transmitter, the steps are as follows.

Follow the steps below to bind in 2-way binding:

1. Select [Antelope 2-way] for RF standard of the transmitter, then put the transmitter into the binding mode.
2. The receiver supports three methods for entering the binding mode: via

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对码 Binding

2. 本接收机支持三种方式进入对码状态：按键对码、对码线对码和通电后按键对码
   - 按键对码：首先按下接收机的对码按键，给接收机通电，此时接收机 LED 灯快闪，表示已进入对码状态。然后松开对码键。
   - 对码线对码：对码线接入对码接口（BVD 接口和 CH3 信号端）后，给接收机通电，接收机 LED 灯快闪表示进入对码状态。注意对码成功后取下对码线。
   - 通电后按键对码：如果接收机通电后尚未与发射机通信，长按对码键 3 秒钟，此时接收机 LED 灯快闪，表示已进入对码状态。然后松开对码键。
3. 当接收机的 LED 灯常亮时，表示对码成功（发射机在对码成功后会自动退出对码状态。）；
4. 检查发射机、接收机是否正常工作。如需重新对码，请重复以上步骤。

单向对码步骤：

1. 发射机选择单向通信，然后进入对码状态；
2. 本接收机进入对码状态（具体进入对码状态的方法请参考双向对码时的描述）；
3. 接收机 LED 灯变为慢闪后将发射机退出对码状态。此时，如果接收机的 LED 灯常亮，表示对码成功；
4. 检查发射机和接收机是否正常工作。如果需要重新对码，请重复上述步骤。

BIND button, via bind cable, or via BIND button after power-on.
- Via BIND button: Press and hold the BIND button of the receiver while powering on the receiver, the LED of the receiver should be flashing, indicating that the receiver is in binding mode. Then release the BIND button.
- Via bind cable: Insert the bind cable into the bind interface (the signal pins of the BVD and CH3 interfaces), then power on the receiver. The LED will flash to indicate binding mode. Remove the bind cable after binding completes.
- Via BIND button after power-on: The receiver is not connected to the transmitter when powered on. Press and hold the BIND button for 3 seconds, the LED should be flashing, indicating that the receiver is in bind mode. Then release the BIND button.

3. When the receiver's LED is solid on, it indicates that the binding is successful (The transmitter will automatically exit the binding state after a successful binding).
4. Check if the transmitter and receiver are working properly. If you need to re-bind, please repeat the above steps.

Follow the steps below to bind in 1-way binding:

1. Select [Antelope 1-way] for RF standard of the transmitter, then put the transmitter into the binding mode.
2. This receiver enters the binding state (for details on how to enter the binding state, refer to the description provided for 2-way binding).
3. After the receiver LED becomes slow flashing, then put the transmitter to exit the binding state. If the receiver's LED is solid on, it indicates that the binding is successful.
4. Check to make sure the transmitter and receiver functions are working correctly, repeat steps above(binding process) if any problems arise.

LED 灯 LED Indicator

接收机状态指示灯用于指示接收机的电源及工作状态。

- LED 红色慢闪：表示未与发射机通信或掉码（失控状态）。
- LED 红色常亮：表示与发射机通信正常。
- LED 红色快闪：表示对码状态。
- LED 红色三闪一灭：表示强制更新状态。
- LED 红色 1 闪 1 长亮 1 长灭：作为 PWM 转换器使用，但未收到有效 i-BUS/i-BUS2 信号。
- LED 红色 1 闪 1 长亮：作为 PWM 转换器使用且收到有效 i-BUS/i-BUS2 信号。

The LED indicates the power and operating status of the receiver.

- The LED flashes red slowly: The receiver is not communicating with the transmitter or is in a signal loss state(out-of-control state).
- The LED is red solid on: The receiver is communicating normally with the transmitter.
- The LED flashes red rapidly: The receiver is in the binding state.
- The LED in red operates in a three-flash-one-off state repeatedly: The receiver firmware is in a forced update state.
- The LED flashes red in the pattern of one short flash, one long solid on, then one long off: The device is operating as a PWM converter, but no valid i-BUS / i-BUS2 signal is received.
- The LED flashes red in the pattern of one short flash followed by one long solid on: The device is operating as a PWM converter and a valid i-BUS / i-BUS2 signal is received.

固件更新 Firmware Update

本接收机的固件更新可使用以下两种方式更新：

方式一：通过发射机更新

使用一台已与接收机对码且内置接收机固件的发射机（如 Noble Mix）进行更新。具体步骤请参阅该发射机的说明书。

方式二：通过富斯遥控管家（FlySkyAssistant）更新

请注意：仅 3.0 及以上版本的富斯遥控管家支持此操作。相关固件可从官网 www.flyskytech.com 下载。此方式下，有两种方法可使接收机进入更新状态：

方法一：先完成发射机与接收机的对码（接收机 LED 灯常亮），然后将发射机连接至电脑。在电脑上打开富斯遥控管家，通过该软件进行固件更新。

方法二（强制更新）：首先将发射机连接至电脑，然后按以下任一方式使接收机进入强制更新状态（此时

The firmware of this receiver can be updated using either of the following two methods:

Method 1 – Via a transmitter

Use a transmitter that has been bound to the receiver and contains the receiver firmware (e.g., Noble Mix). Refer to the transmitter's manual for detailed steps.

Method 2 – Via FlySkyAssistant

Note: Only version 3.0 or later of FlySkyAssistant supports this operation. The firmware can be downloaded from www.flysky-cn.com. Two ways are available under this method:

Way1: Bind the transmitter to the receiver (receiver LED stays solid on), then connect the transmitter to a PC. Open FlySkyAssistant on the PC and update the firmware via the software.

Way2 (Forced update): Connect the transmitter to the PC first, then use one of the following methods to force the receiver into update mode

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固件更新 Firmware Update

接收机 LED 指示灯为 3 闪 1 灭）：

- 按住对码按键同时给接收机通电，等待约 10 秒直至指示灯变为 3 闪 1 灭，然后松开对码按键。
- 或先给接收机通电，然后长按对码键 10 秒，直到指示灯 3 闪 1 灭，随后松开对码按键。
- 或将对码线接入 CH1 和 CH3 信号端，然后给接收机通电，指示灯 3 闪 1 灭。

完成上述任一操作步骤后，在电脑端打开富斯遥控管家软件，通过该软件完成固件的强制更新。强制更新完成后，接收机的指示灯将由 3 闪 1 灭状态变为慢闪状态。

(indicated by a "three-flash-one-off " LED pattern):

- Press and hold the bind button while powering on the receiver. Wait approximately 10 seconds until the LED operates in a three-flash-one-off pattern repeatedly, then release the BIND button.
- Power on the receiver first, then press and hold the BIND button for about 10 seconds until the LED operates in a three-flash-one-off pattern repeatedly, then release the BIND button.
- Short the signal pins of CH1 and CH3 interfaces using a bind cable, then power on the receiver. Thne the LED operates in a three-flash-one-off pattern repeatedly.

After any of the above steps is completed, open FlySkyAssistant on your computer and complete the forced firmware update through the software. Once the forced update is completed, the LED of the receiver will change from the three-flash-one-off state to a slow flashing state.

失控保护 Failsafe

失控保护功能用于在接收机失去信号不受控制后，接收机按设置好的失控保护值进行通道输出以保护模型及人员安全。

本款接收机共支持三种失控保护模式 : [无输出]、[保持] 和 [固定值]

[无输出]PWM 通道接口为无输出状态 ;

[保持] 输出失控前最后的通道值;

[固定值] 输出设置的通道值。

注：

1. 对于 PPM/i-BUS/S.BUS/i-BUS2 等总线信号类型不允许单个或其中几个通道为 [无输出] 模式，通道设置为 [无输出] 模式时，实际信号是保持最后输出值；
2. 因 S.BUS/i-BUS2 信号信息包含失控标志位，各通道失控保护设置被失控标志位传达给后续设备，若连接的设备支持失控标志位解析，则失控后，输出各通道设置的失控保护值；
3. 对于无失控标志位的信号 PPM/i-BUS，支持设置失控时信号 [无输出] 模式。设置为 [无输出] 模式后，不管各通道失控保护如何设置，失控后各通道均为 [无输出] 模式。

The failsafe function is used when the receiver loses RF signal and is out-of-control. The receiver performs channel output according to the set failsafe value to protect the safety of the model and personnel.

This receiver supports three fail-safe modes: **No output**, **Hold**, and **Fixed value**.

No output: No output for PWM channel.

Hold: Maintain the last output value.

Fixed value : Output the failsafe values that have been set for each channel.

Notes:

1. For bus signal types such as PPM/i-BUS/S.BUS/i-BUS2, a single or several of these channels are not allowed to be in No output mode. The actual signal is held at the last output value when the channel is set to No output mode.
2. Because the S.BUS/i-BUS2 signal information contains failsafe flag bits, the failsafe settings of each channel are communicated to subsequent devices by the failsafe flag bits. If the connected devices support the failsafe flag bit analysis, the failsafe values set for each channel are output after out of control.
3. For the signal PPM/i-BUS without failsafe flag bits, it supports the setting of the signal to No output mode in case of out of control. After setting to No output mode, regardless of the setting of the failsafe of each channel, each channel will be in No output mode after out-of-control.

其他功能 Other Features

本接收机与发射机双向对码完成后，支持遥测回传。

它也可配置为 i-BUS / i-BUS2 协议的 PWM 转换器，用于扩展通道数量。

此外，接收机还支持 RSSI 输出与 BVD 电池电压检测功能。

相关功能设置请参考对应发射机的说明书。

After the receiver and transmitter are successfully bound via 2-way binding, telemetry becomes available.

The receiver may also be configured as a PWM converter using the i-BUS or i-BUS2 protocol to increase channel capacity.

Additionally, the receiver supports RSSI output and BVD (Battery Voltage Detection).

Please refer to the corresponding transmitter manual for related function settings.

注意事项：

- 使用前必须确保本产品与模型安装正确，否则可能导致模型发生严重损坏。
- 为了一切正常，请养成先开发射机再接收机通电以及先接收机断电再关闭发射机的习惯。
- 确保接收机安装在远离电机，电子调速器或电子噪声过多的区域。
- 接收机天线需远离导电材料，例如金属棒和碳物质。为了避免影响正常工作，请确保接收机天线和导电材料之间至少有 1 厘米以上的距离。
- 准备过程中，请勿连接接收机电源，避免造成不必要的损失。

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Attention:

- Make sure the product is installed and calibrated correctly, failure to do so may result in serious injury.
- Normally, you must power on the transmitter and then receiver, and power off the receiver and then the transmitter.
- Make sure the receiver is mounted away from motors, electronic speed controllers or any device that emits excessive electrical noise.
- Keep the receiver's antenna at least 1cm away from conductive materials such as carbon or metal.
- Do not power on the receiver during the setup process to prevent loss of control.

认证相关 Certification

FCC Compliance Statement

This device complies with Part 15 of the FCC Rules. Operation is subject to the following two conditions: (1) this device may not cause harmful interference, and (2) this device must accept any interference received, including interference that may cause undesired operation.

Warning: changes or modifications not expressly approved by the party responsible for compliance could void the user's authority to operate the equipment.

This equipment has been tested and found to comply with the limits for a Class B digital device, pursuant to Part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference in a residential installation. This equipment generates, uses and can radiate radio frequency energy and, if not installed and used in accordance with the instructions, may cause harmful interference to radio communications. However, there is no guarantee that interference will not occur in a particular installation.
If this equipment does cause harmful interference to radio or television reception, which can be determined by turning the equipment off and on, the user is encouraged to try to correct the interference by one or more of the following measures:
-- Reorient or relocate the receiving antenna.
-- Increase the separation between the equipment and receiver.
-- Connect the equipment into an outlet on a circuit different from that to which the receiver is connected.
-- Consult the dealer or an experienced radio/TV technician for help.

EU DoC Declaration

Hereby, [ShenZhen FLYSKY Technology Co., Ltd.] declares that the Radio Equipment [Ar4] is in compliance with RED 2014/53/EU.
The full text of the EU DoC is available at the following internet address: www.flyskytech.com/info_detail/10.html

RF Exposure Compliance

This equipment complies with FCC/ISED RF radiation exposure limits set forth for an uncontrolled environment. This equipment should be installed and operated with a minimum distance of 20 centimeters between the radiator and your body.

Environmentally friendly disposal

Old electrical appliances must not be disposed of together with the residual waste, but have to be disposed of separately. The disposal at the communal collecting point via private persons is for free. The owner of old appliances is responsible to bring the appliances to these collecting points or to similar collection points. With this little personal effort, you contribute to recycle valuable raw materials and the treatment of toxic substances.

FCC ID: 2A2UNAR400

微信公众号 Bilibili Website Facebook

Manufacturer: ShenZhen FLYSKY Technology Co., Ltd.
Address: 16F, Huafeng Building, No. 6006 Shennan Road, Futian District, Shenzhen, Guangdong, China

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Figures and illustrations in this manual are provided for reference only and may differ from actual product appearance. Product design and specificatiions may be changed without notice.